Course: CS 3650 - Human-Centered Design

Target System: Intermountain Health / MyChart

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GitHub Repository: https://github.com/TannerMcG/CS3650-ContextResearch-DiagnosticUsabilityAudit.git

1. System Selection & Problem Statement

1.1 Context of Use

- **Platform Name:** Intermountain Health MyChart
- **Target User Demographics:** Adult humans across the state of Utah looking to access the medical history, make future appointments, message their doctor, make payments on medical bills.
- **Core User Tasks:**
    1. Make Appointments: find a primary care doctor/specialist and make an appointment with them.
    2. Message doctors: Have a easy messaging systems to contact patients doctors with questions or needing to get prescriptions.
    3. Medical Records: Have a database with users past medical history and labs.

1.2 Problem Statement

The Intermountain Health MyChart patient portal is a place for Utahns that use Intermountain Health for any of their medical needs. Users would be able to access their medical history, make appointments, message their doctor directly, or make payments on their medical bills. When taking a look at the MyChart portal UI there are a few UX elements that can lead to patient frustration, task abandonment, and an increased reliance on making phone calls instead of doing the tasks yourself.

2. Qualitative Contextual Research

2.1 User Research Protocol

- **Warm-up:** How frequently do you log into you MyChart patent portal and what is your main goal for logging on?
- **Task 1 (Scheduling):** If you needed to make an appointment next week with your primary doctor what steps would you take to do that?

- **Task 2 (Messaging):** Say your medicine you are taking is not enough and you need a higher prescription, how would you go about contacting your doctor to increase that?
- **Task 3 (Lab Results):** If you recently had lab work done at the hospital, how would you find those results in the patient portal?

2.2 Qualitative Findings & Synthesis

User / Profile	**Task Observed**	**Mental Model Mismatch / Pain Point**	**Direct User Quote**
User 1(Age 25, frequent user)	Task 1: Scheduling	They expected a screen to select what doctor they wanted to see and then their schedule of availability. They thought it was simple and easy to do	"Everything is pretty basic, I choose my doctor then a time."
User 2(Age 53, experienced user	Task2: Messaging	They said they have messaged their doctor before on there to refill prescriptions so they knew what they were doing. They didn't like that in the messaging it says that "Your care team may use artificial intelligence recommendations" and felt it wouldn't be as accurate as a normal doctor.	"Why would I message them if they are just going to have an AI respond instead of the professional."
User 3(Age 20, non-frequent user)	Task 3: Lab Results	They had never opened the MyChart page. They assumed their resent lab	"What is Hemoglobin A1C."

User / Profile	Task Observed	Mental Model Mismatch / Pain Point	Direct User Quote
		results would just show up on the homepage. When looking at the lab results they did not understand what any of the medical words meant so they didn't know where to find a specific result	

3. Heuristic Evaluation & WCAG 2.1 AA Audit

3.1 Nielsen's 10 Usability Heuristics Audit

(Severity Ratings: 0 = None, 1 = Cosmetic, 2 = Minor, 3 = Major, 4 = Catastrophe)

Heuristic Principle	Observed Flaw & UI Location	Severity Rating
1. Visibility of System Status	Clicking on the "Schedule Now" action button pulled up a active loading screen with "gathering scheduling options"	0
2. Match Between System & Real World	After clicking onto the test results page a lot of medical terms for what test have been done show which users would not know what those mean.	2
3. Consistency & Standards	When going through multiple layers on the UI in the Update Contact Info page the only button that appears to stop is "Discard changes" which does say it will go back but most users would imply that	1

Heuristic Principle	Observed Flaw & UI Location	Severity Rating
4. Error Prevention	When going to different pages on the website the layout stays with the top bar format being the same and predictable	0
5. Error Prevention	After testing the personal information and changing details about me there was no error prevention to stop me from making a mistake	3
6. Recognition Rather Than Recall	The system does not use any recognition of things that might have been input on one screen from another so users would have to recall from their memory	3
7. Flexibility & Efficiency of Use	There is some customization ability that can be done on the main page by pinning your 6 most used items, but only six no more and no less.	2
8. Aesthetic & Minimalist Design	The overall design has minimal clutter but some elements pop more than others like the “Your Updates” section takes away from the top 6 pages above it with those being smaller	1
9. Help Users Recognize & Recover from Errors	When waiting for the session expired a popup showed up showing that the user would be logged out and have to log back in	0
10. Help & Documentation	When users get stuck there is a small chat box in the bottom righthand corner but it just shows a logo of Intermountain Health and	1

Heuristic Principle	Observed Flaw & UI Location	Severity Rating
	nothing about “help”, but when clicking on it, it asks how it can help	

3.2 Baseline Accessibility (WCAG 2.1 AA) Audit

WCAG 2.1 Criteria	Test Method & Observation	Compliance Status
SC 1.4.3 Contrast (Minimum)	Using the DevTools and scanning the MyChart dashboard found that there weren’t any violations of the 4.5:1 color contrast.	Compliant
SC 2.4.7 Focus Visible	From doing manual keyboard navigation using the Tab key, I was able to confirm that the interactive parts of the homepage were visible throughout.	Compliant
SC 2.4.4 / 4.1.2 Link Purpose & Accessible Names	The scan using DevTools showed that certain interactive icons and buttons lack discernible text names for screen readers	Non-Compliant
SC 2.5.5 / 2.5.8 Touch Target Size & Spacing	The scan using DevTools flagged some instances where interactive touch targets fall below the recommended minimum sizing and spacing for those users.	Non-Compliant

4. Post-Mortem Reflection (Metacognition)

Question 1: Expert Audits vs. Qualitative User Observations

The way that I was able to see some usability flaws that a developer testing their own code would likely overlook was seeing what basic functionality do users expect. Code

developers can get so caught up in the things they make that they forget the easy things most people would expect.

Question 2: Accessibility Violations & Universal Design

A place where I was able to identify some friction during the audit was the interactive icon buttons and links were lacking discernible text names. This violated the WCAG SC 4.1.2 Name, Role, Value and SC 2.4.4 Link Purpose. Those are about when the screen reader is announcing things to users that need it, it would label the button for instance as just “button” which leaves the user not knowing what that button would do. If they were able to resolve that violation then it would improve the user experience a lot for disabled users and for the general public because that could help elderly people find/do what they need and their caretakers would not have to do it for them.

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